Concept Note

Breast Cancer Prediction using Machine Learning | Kamini Shanmukha | Ideal Institute of Technology, Kakinada | AICTE | IBM SkillsBuild — ML & Applied AI Internship 2026

■ SDG 3: Good Health and Well-being

This project aligns with UN Sustainable Development Goal 3, which calls for ensuring healthy lives and promoting well-being for all. Early and accurate detection of disease is central to this goal, and machine learning can meaningfully support that effort.

AI Project Ideas & Relevance

- AI/ML-based early disease detection using diagnostic measurements from medical data.
- A classification model that flags likely-malignant cell patterns to support faster, more consistent preliminary screening.

Problem Statement

> "Manual interpretation of biopsy and cell-measurement data for breast cancer diagnosis is time-consuming and can be inconsistent, and limited access to specialist pathologists in smaller hospitals delays early detection. This project uses machine learning to classify tumor characteristics as malignant or benign, enabling faster, more consistent, and more accessible preliminary screening."

Proposed Solution & Approach

- Use the Breast Cancer Wisconsin (Diagnostic) dataset — 569 samples, 30 numeric features describing cell nuclei characteristics.
- Perform exploratory data analysis to understand class balance and feature correlations.
- Train and compare three classification models: Logistic Regression, Random Forest, and SVM, using cross-validation.
- Tune the best-performing model with GridSearchCV to optimize its hyperparameters.
- Evaluate the final model using accuracy, precision, recall, F1-score, and ROC-AUC — prioritizing recall, since missing a malignant case is more costly than a false alarm in a screening context.

Expected Outcome & Impact

A trained machine learning model capable of classifying tumors as malignant or benign with high accuracy and recall, demonstrating how AI can support — not replace — clinical decision-making. In practice, such a tool could help diagnostic labs and smaller hospitals provide faster preliminary screening, supporting earlier treatment and better patient outcomes.

Tools & Technologies

Python, scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Google Colab.

Lean Canvas

AI-Based Breast Cancer Prediction | SDG 3: Good Health & Well-being

PROBLEM

List your top 1-3 problems

- Manual interpretation of biopsy/cell data for cancer diagnosis is time-consuming and can be inconsistent.
- Limited access to specialist pathologists/oncologists in smaller hospitals delays early detection.
- Late detection significantly reduces survival rates and treatment options.

SOLUTION

Outline a possible solution for each problem

- ML model that classifies tumor characteristics (cell nuclei measurements) as malignant or benign.
- Acts as a fast, consistent second opinion supporting - not replacing - clinical diagnosis.

UNIQUE VALUE PROPOSITION

Single, clear, compelling message

- Faster, more consistent preliminary cancer screening powered by machine learning - supporting doctors, not replacing them.

UNFAIR ADVANTAGE

Something that cannot easily be bought or copied

- Rigorous, reproducible ML methodology: cross-validation, hyperparameter tuning, and evaluation beyond accuracy (recall, ROC-AUC) tuned for medical screening.

CUSTOMER SEGMENTS

Target customers and users

- Diagnostic labs and hospitals, especially in under-resourced areas.
- Pathologists and oncologists seeking a fast preliminary read.

EXISTING ALTERNATIVES

How these problems are solved today

- Manual microscopic examination by pathologists.
- Mammography/biopsy review relying solely on human expert judgment.

KEY METRICS

Numbers that tell you how it's doing

- Model accuracy, precision, recall, F1-score, ROC-AUC on test data.
- Reduction in average time to preliminary diagnosis.

HIGH-LEVEL CONCEPT

Your X for Y analogy

- Like a spell-checker, but for flagging likely-malignant cell patterns before specialist review.

CHANNELS

Path to customers

- Direct outreach to hospitals and diagnostic labs.
- Integration with existing lab/hospital software.
- Healthcare-AI platforms and academic conferences.

EARLY ADOPTERS

Characteristics of your ideal customer

- Smaller clinics/labs without in-house specialist pathologists.

COST STRUCTURE

Fixed and variable costs

- Data acquisition/licensing and computing resources for training.
- Model maintenance, monitoring, and periodic retraining.
- Regulatory and compliance costs for clinical use.

REVENUE STREAMS

Sources of revenue

- Subscription/licensing fee per lab or hospital.
- Pay-per-screening usage-based pricing.